

Rational Numbers

Exercise 9.1

Question 1:

List five rational numbers between:

(i) -1 and 0 (ii) -2 and -1

(iii) $\frac{-4}{5}$ and $\frac{-2}{3}$ (iv) $\frac{1}{2}$ and $\frac{2}{3}$

Answer:

(i) -1 and 0

$$\frac{-1}{10}, \frac{-1}{20}, \frac{-1}{30}, \frac{-1}{40}, \frac{-1}{50}$$

(ii) -2 and -1

$$-2 = \frac{-12}{6} \text{ and } -1 = \frac{-6}{6}$$

Five rational numbers are

$$\frac{-11}{6}, \frac{-10}{6}, \frac{-9}{6}, \frac{-8}{6}, \frac{-7}{6}$$

(iii) $\frac{-4}{5}$ and $\frac{-2}{3}$

$$\frac{-4}{5} = \frac{-4 \times 9}{5 \times 9} = \frac{-36}{45} \text{ and } \frac{-2}{3} = \frac{-2 \times 15}{3 \times 15} = \frac{-30}{45}$$

Five rational numbers are

$$\frac{-35}{45}, \frac{-34}{45}, \frac{-33}{45}, \frac{-32}{45}, \frac{-31}{45}$$

$$(iv) \frac{1}{2} \text{ and } \frac{2}{3}$$

$$\frac{1}{2} = \frac{1 \times 18}{2 \times 18} = \frac{18}{36} \text{ and } \frac{2}{3} = \frac{2 \times 12}{3 \times 12} = \frac{24}{36}$$

Five rational numbers are

$$\frac{19}{36}, \frac{20}{36}, \frac{21}{36}, \frac{22}{36}, \frac{23}{36}$$

Question 2:

Write four more rational numbers in each of the following patterns:

$$(i) \frac{-3}{5}, \frac{-6}{10}, \frac{-9}{15}, \frac{-12}{20}, \dots \quad (ii) \frac{-1}{4}, \frac{-2}{8}, \frac{-3}{12}, \dots$$

$$(iii) \frac{-1}{6}, \frac{2}{-12}, \frac{3}{-18}, \frac{4}{-24}, \dots \quad (iv) \frac{-2}{3}, \frac{2}{-3}, \frac{4}{-6}, \frac{6}{-9}, \dots$$

Answer:

$$(i) \frac{-3}{5}, \frac{-6}{10}, \frac{-9}{15}, \frac{-12}{20}, \dots$$

$$\frac{-3}{5}, \frac{-3 \times 2}{5 \times 2}, \frac{-3 \times 3}{5 \times 3}, \frac{-3 \times 4}{5 \times 4}, \dots$$

It can be observed that the numerator is a multiple of 3 while the denominator is a multiple of 5 and as we increase them further, these multiples are increasing.

Therefore, the next four rational numbers in this pattern are

$$\frac{-3 \times 5}{5 \times 5}, \frac{-3 \times 6}{5 \times 6}, \frac{-3 \times 7}{5 \times 7}, \frac{-3 \times 8}{5 \times 8}, \dots$$

$$\frac{-15}{25}, \frac{-18}{30}, \frac{-21}{35}, \frac{-24}{40}, \dots$$

(ii)

$$\frac{-1}{4}, \frac{-2}{8}, \frac{-3}{12} \dots$$

$$\frac{-1}{4}, \frac{-1 \times 2}{4 \times 2}, \frac{-1 \times 3}{4 \times 3} \dots$$

The next four rational numbers in this pattern are

$$\frac{-1 \times 4}{4 \times 4}, \frac{-1 \times 5}{4 \times 5}, \frac{-1 \times 6}{4 \times 6}, \frac{-1 \times 7}{4 \times 7} \dots$$

$$\frac{-4}{16}, \frac{-5}{20}, \frac{-6}{24}, \frac{-7}{28} \dots$$

(iii)

$$\frac{-1}{6}, \frac{2}{-12}, \frac{3}{-18}, \frac{4}{-24} \dots$$

$$\frac{-1}{6}, \frac{1 \times 2}{-6 \times 2}, \frac{1 \times 3}{-6 \times 3}, \frac{1 \times 4}{-6 \times 4} \dots$$

The next four rational numbers in this pattern are

$$\frac{1 \times 5}{-6 \times 5}, \frac{1 \times 6}{-6 \times 6}, \frac{1 \times 7}{-6 \times 7}, \frac{1 \times 8}{-6 \times 8} \dots$$

$$\frac{5}{-30}, \frac{6}{-36}, \frac{7}{-42}, \frac{8}{-48} \dots$$

(iv) $\frac{-2}{3}, \frac{2}{-3}, \frac{4}{-6}, \frac{6}{-9} \dots$

$$\frac{-2}{3}, \frac{2}{-3}, \frac{2 \times 2}{-3 \times 2}, \frac{2 \times 3}{-3 \times 3} \dots$$

The next four rational numbers in this pattern are

$$\frac{2 \times 4}{-3 \times 4}, \frac{2 \times 5}{-3 \times 5}, \frac{2 \times 6}{-3 \times 6}, \frac{2 \times 7}{-3 \times 7} \dots$$

$$\frac{8}{-12}, \frac{10}{-15}, \frac{12}{-18}, \frac{14}{-21} \dots$$

Question 3:

Give four rational numbers equivalent to:

(i) $\frac{-2}{7}$ (ii) $\frac{5}{-3}$ (iii) $\frac{4}{9}$

Answer:

(i) $\frac{-2}{7}$

Four rational numbers are

$$\frac{-2 \times 2}{7 \times 2}, \frac{-2 \times 3}{7 \times 3}, \frac{-2 \times 4}{7 \times 4}, \frac{-2 \times 5}{7 \times 5}$$
$$\frac{-4}{14}, \frac{-6}{21}, \frac{-8}{28}, \frac{-10}{35}$$

(ii) $\frac{5}{-3}$

Four rational numbers are

$$\frac{5 \times 2}{-3 \times 2}, \frac{5 \times 3}{-3 \times 3}, \frac{5 \times 4}{-3 \times 4}, \frac{5 \times 5}{-3 \times 5}$$
$$\frac{10}{-6}, \frac{15}{-9}, \frac{20}{-12}, \frac{25}{-15}$$

(iii) $\frac{4}{9}$

Four rational numbers are

$$\frac{4 \times 2}{9 \times 2}, \frac{4 \times 3}{9 \times 3}, \frac{4 \times 4}{9 \times 4}, \frac{4 \times 5}{9 \times 5}$$
$$\frac{8}{18}, \frac{12}{27}, \frac{16}{36}, \frac{20}{45}$$

Question 4:

Draw the number line and represent the following rational numbers on it:

(i) $\frac{3}{4}$ (ii) $\frac{-5}{8}$

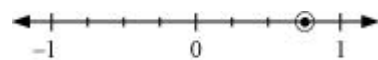
(iii) $\frac{-7}{4}$ (iv) $\frac{7}{8}$

Answer:

(i) $\frac{3}{4}$

This fraction represents 3 parts out of 4 equal parts. Therefore, each space between two integers on number line must be divided into 4 equal parts.

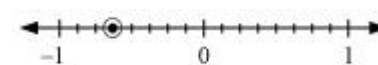
$\frac{3}{4}$ can be represented as



(ii) $\frac{-5}{8}$

This fraction represents 5 parts out of 8 equal parts. Negative sign represents that it is on the negative side of number line. Therefore, each space between two integers on number line must be divided into 8 equal parts.

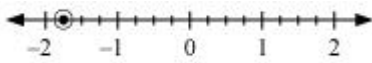
$\frac{-5}{8}$ can be represented as



(iii) $\frac{-7}{4} = -1\frac{3}{4}$

This fraction represents 1 full part and 3 parts out of 4 equal parts. Negative sign represents that it is on the negative side of number line. Therefore, each space between two integers on number line must be divided into 4 equal parts.

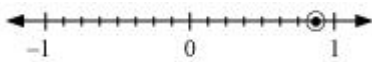
$\frac{-7}{4}$ can be represented as



(iv) $\frac{7}{8}$

This fraction represents 7 parts out of 8 equal parts. Therefore, each space between two integers on number line must be divided into 8 equal parts.

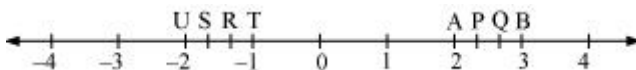
$\frac{7}{8}$ can be represented as



Question 5:

The points P, Q, R, S, T, U, A and B on the number line are such that,

$TR = RS = SU$ and $AP = PQ = QB$. Name the rational numbers represented by P, Q, R and S.



Answer:

Distance between U and T = 1 unit

It is divided into 3 equal parts.

$$TR = RS = SU = \frac{1}{3}$$

$$R = -1 - \frac{1}{3} = -\frac{3}{3} - \frac{1}{3} = \frac{-4}{3}$$

$$S = -1 - \frac{2}{3} = -\frac{3}{3} - \frac{2}{3} = -\frac{5}{3}$$

Similarly,

$$AB = 1 \text{ unit}$$

It is divided into 3 equal parts.

$$P = 2 + \frac{1}{3} = \frac{6}{3} + \frac{1}{3} = \frac{7}{3}$$

$$Q = 2 + \frac{2}{3} = \frac{6}{3} + \frac{2}{3} = \frac{8}{3}$$

Question 6:

Which of the following pairs represent the same rational number?

$$(i) \frac{-7}{21} \text{ and } \frac{3}{9} \quad (ii) \frac{-16}{20} \text{ and } \frac{20}{-25} \quad (iii) \frac{-2}{-3} \text{ and } \frac{2}{3}$$

$$(iv) \frac{-3}{5} \text{ and } \frac{-12}{20} \quad (v) \frac{8}{-5} \text{ and } \frac{-24}{15} \quad (vi) \frac{1}{3} \text{ and } \frac{-1}{9}$$

$$(vii) \frac{-5}{-9} \text{ and } \frac{5}{-9}$$

Answer:

$$(i) \frac{-7}{21} \text{ and } \frac{3}{9}$$

$$\frac{-7}{21} = \frac{-1}{3}$$

$$\frac{3}{9} = \frac{1}{3}$$

As $\frac{-1}{3} \neq \frac{1}{3}$, therefore, it does not represent same rational numbers.

(ii) $\frac{-16}{20}$ and $\frac{20}{-25}$

$$\frac{-16}{20} = \frac{-4}{5}$$

$$\frac{-20}{25} = \frac{-4}{5}$$

Therefore, it represents same rational numbers.

(iii) $\frac{-2}{-3}$ and $\frac{2}{3}$

$$\frac{-2}{-3} = \frac{2}{3}$$

Therefore, it represents same rational numbers.

(iv) $\frac{-3}{5}$ and $\frac{-12}{20}$

$$\frac{-12}{20} = \frac{-3}{5}$$

Therefore, it represents same rational numbers.

(v) $\frac{8}{-5}$ and $\frac{-24}{15}$

$$\frac{-24}{15} = \frac{-8}{5}$$

$$\frac{8}{-5} = \frac{-8}{5}$$

Therefore, it represents same rational numbers.

(vi) $\frac{1}{3}$ and $\frac{-1}{9}$

As $\frac{1}{3} \neq \frac{-1}{9}$, therefore, it does not represent same rational numbers.

(vii) $\frac{-5}{-9}$ and $\frac{5}{-9}$

$$\frac{-5}{-9} = \frac{5}{9}$$

As $\frac{5}{9} \neq \frac{-5}{9}$, therefore, it does not represent same rational numbers.

Question 7:

Rewrite the following rational numbers in the simplest form:

(i) $\frac{-8}{6}$ (ii) $\frac{25}{45}$

(iii) $\frac{-44}{72}$ (iv) $\frac{-8}{10}$

Answer:

(i) $\frac{-8}{6} = \frac{-4 \times 2}{3 \times 2} = \frac{-4}{3}$

(ii) $\frac{25}{45} = \frac{5 \times 5}{9 \times 5} = \frac{5}{9}$

(iii) $\frac{-44}{72} = \frac{-11 \times 4}{18 \times 4} = \frac{-11}{18}$

(iv) $\frac{-8}{10} = \frac{-4 \times 2}{5 \times 2} = \frac{-4}{5}$

Question 8:

Fill in the boxes with the correct symbol out of >, <, and =

$$(i) \frac{-5}{7} \square \frac{2}{3} \quad (ii) \frac{-4}{5} \square \frac{-5}{7} \quad (iii) \frac{-7}{8} \square \frac{14}{-16}$$

$$(iv) \frac{-8}{5} \square \frac{-7}{4} \quad (v) \frac{1}{-3} \square \frac{-1}{4} \quad (vi) \frac{5}{-11} \square \frac{-5}{11}$$

$$(vii) 0 \square \frac{-7}{6}$$

Answer:

(i)

$$\frac{-5}{7} = \frac{-5 \times 3}{7 \times 3} = \frac{-15}{21}$$

$$\frac{2}{3} = \frac{2 \times 7}{3 \times 7} = \frac{14}{21}$$

As $-15 < 14$,

Therefore, $\frac{-5}{7} \square \frac{2}{3}$

(ii)

$$\frac{-4}{5} = \frac{-4 \times 7}{5 \times 7} = \frac{-28}{35}$$

$$\frac{-5}{7} = \frac{-5 \times 5}{7 \times 5} = \frac{-25}{35}$$

As $-28 < -25$

Therefore, $\frac{-4}{5} \square \frac{-5}{7}$

(iii) Here, $\frac{14}{-16} = \frac{7 \times 2}{-8 \times 2} = \frac{7}{-8} = \frac{-7}{8}$

Therefore, $\frac{-7}{8} \square \frac{14}{-16}$

(iv)

$$\frac{-8}{5} = \frac{-8 \times 4}{5 \times 4} = \frac{-32}{20}$$

$$\frac{-7}{4} = \frac{-7 \times 5}{4 \times 5} = \frac{-35}{20}$$

As $-32 > -35$,

Therefore, $\frac{-8}{5} \boxed{>} \frac{-7}{4}$

(v)

$$\frac{-1}{3} = \frac{-1 \times 4}{3 \times 4} = \frac{-4}{12}$$

$$\frac{-1}{4} = \frac{-1 \times 3}{4 \times 3} = \frac{-3}{12}$$

As $-4 < -3$,

Therefore, $\frac{-1}{3} \boxed{<} \frac{-1}{4}$

(vi) $\frac{5}{-11} \boxed{=} \frac{-5}{11}$

(vii) $0 \boxed{>} \frac{-7}{6}$

Question 9:

Which is greater in each of the following?

(i) $\frac{2}{3}, \frac{5}{2}$ (ii) $\frac{-5}{6}, \frac{-4}{3}$ (iii) $\frac{-3}{4}, \frac{2}{-3}$

$$(iv) \frac{-1}{4}, \frac{1}{4} \quad (v) -3\frac{2}{7}, -3\frac{4}{5}$$

Answer:

$$(i) \frac{2}{3}, \frac{5}{2}$$

By converting these into like fractions,

$$\frac{2}{3} = \frac{2 \times 2}{3 \times 2} = \frac{4}{6}$$

$$\frac{5}{2} = \frac{5 \times 3}{2 \times 3} = \frac{15}{6}$$

As $15 > 4$, therefore, $\frac{5}{2}$ is greater.

$$(ii) \frac{-5}{6}, \frac{-4}{3}$$

$$\frac{-4}{3} = \frac{-4 \times 2}{3 \times 2} = \frac{-8}{6}$$

As $-5 > -8$, therefore, $\frac{-5}{6}$ is greater.

(iii)

$$\frac{-3}{4}, \frac{2}{-3}$$

$$\text{Or, } \frac{-3}{4}, \frac{-2}{3}$$

By converting these into like fractions,

$$\frac{-3}{4} = \frac{-3 \times 3}{4 \times 3} = \frac{-9}{12}$$

$$\frac{-2}{3} = \frac{-2 \times 4}{3 \times 4} = \frac{-8}{12}$$

As $-8 > -9$, therefore, $\frac{-2}{3}$ is greater.

$$(iv) \quad \frac{-1}{4}, \frac{1}{4}$$

$$\frac{1}{4} > \frac{-1}{4}$$

$$(v) \quad -3\frac{2}{7}, -3\frac{4}{5}$$

$$\frac{-23}{7}, \frac{-19}{5}$$

By converting these into like fractions,

$$\frac{-23}{7} = \frac{-23 \times 5}{7 \times 5} = \frac{-115}{35}$$

$$\frac{-19}{5} = \frac{-19 \times 7}{5 \times 7} = \frac{-133}{35}$$

As $-115 > -133$, therefore, $-3\frac{2}{7}$ is greater.

Question 10:

Write the following rational numbers in ascending order:

$$(i) \quad \frac{-3}{5}, \frac{-2}{5}, \frac{-1}{5} \quad (ii) \quad \frac{-1}{3}, \frac{-2}{9}, \frac{-4}{3} \quad (iii) \quad \frac{-3}{7}, \frac{-3}{2}, \frac{-3}{4}$$

Answer:

$$(i) \quad \frac{-3}{5}, \frac{-2}{5}, \frac{-1}{5}$$

As $-3 < -2 < -1$,

$$\therefore \frac{-3}{5} < \frac{-2}{5} < \frac{-1}{5}$$

$$(ii) \quad \frac{-1}{3}, \frac{-2}{9}, \frac{-4}{3}$$

By converting these into like fractions,

$$\frac{-1 \times 3}{3 \times 3}, \frac{-2}{9}, \frac{-4 \times 3}{3 \times 3}$$

$$\frac{-3}{9}, \frac{-2}{9}, \frac{-12}{9}$$

As $-12 < -3 < -2$,

$$\therefore \frac{-4}{3} < \frac{-1}{3} < \frac{-2}{9}$$

(iii) $\frac{-3}{7}, \frac{-3}{2}, \frac{-3}{4}$

By converting these into like fractions,

$$\frac{-3 \times 4}{7 \times 4}, \frac{-3 \times 14}{2 \times 14}, \frac{-3 \times 7}{4 \times 7}$$

$$\frac{-12}{28}, \frac{-42}{28}, \frac{-21}{28}$$

As $-42 < -21 < -12$,

$$\therefore \frac{-3}{2} < \frac{-3}{4} < \frac{-3}{7}$$

Exercise 9.2

Question 1:

Find the sum:

(i) $\frac{4}{5} + \left(\frac{-11}{4}\right)$ (ii) $\frac{5}{3} + \frac{3}{5}$ (iii) $\frac{-9}{10} + \frac{22}{15}$

(iv) $\frac{-3}{-11} + \frac{5}{9}$ (v) $\frac{-8}{19} + \frac{(-2)}{57}$ (vi) $\frac{-2}{3} + 0$

(vii) $-2\frac{1}{3} + 4\frac{3}{5}$

Answer:

(i) $(\frac{5}{4}) + (-\frac{11}{4})$

We have:

$$= (\frac{5}{4}) - (\frac{11}{4})$$

$$= [(5 - 11)/4] \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (-6/4)$$

$$= -3/2 \dots [\because \text{Divide both numerator and denominator by 3}]$$

(ii) $(\frac{5}{3}) + (\frac{3}{5})$

Take the LCM of the denominators of the given rational numbers.

The LCM of 3 and 5 is 15

Express each of the given rational numbers with the above LCM as the common denominator.

Now,

$$(\frac{5}{3}) = [(5 \times 5) / (3 \times 5)] = (25/15)$$

$$(\frac{3}{5}) = [(3 \times 3) / (5 \times 3)] = (9/15)$$

Then,

$$= (25/15) + (9/15) \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (25 + 9)/15$$

$$= 34/15$$

(iii) $(-\frac{9}{10}) + (\frac{22}{15})$

Take the LCM of the denominators of the given rational numbers.

The LCM of 10 and 15 is 30

Express each of the given rational numbers with the above LCM as the common denominator.

Now,

$$(-9/10) = [(-9 \times 3) / (10 \times 3)] = (-27/30)$$

$$(22/15) = [(22 \times 2) / (15 \times 2)] = (44/30)$$

Then,

$$= (-27/30) + (44/30) \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (-27 + 44)/30$$

$$= (17/30)$$

$$\textbf{(iv) } (-3/-11) + (5/9)$$

We have,

$$= 3/11 + 5/9$$

Take the LCM of the denominators of the given rational numbers.

The LCM of 11 and 9 is 99

Express each of the given rational numbers with the above LCM as the common denominator.

Now,

$$(3/11) = [(3 \times 9) / (11 \times 9)] = (27/99)$$

$$(5/9) = [(5 \times 11) / (9 \times 11)] = (55/99)$$

Then,

$$= (27/99) + (55/99) \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (27 + 55)/99$$

$$= (82/99)$$

$$\textbf{(v) } (-8/19) + (-2/57)$$

We have

$$= -8/19 - 2/57$$

Take the LCM of the denominators of the given rational numbers.

The LCM of 19 and 57 is 57

Express each of the given rational numbers with the above LCM as the common denominator.

Now,

$$(-8/19) = [(-8 \times 3) / (19 \times 3)] = (-24/57)$$

$$(-2/57) = [(-2 \times 1) / (57 \times 1)] = (-2/57)$$

Then,

$$= (-24/57) - (2/57) \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (-24 - 2)/57$$

$$= (-26/57)$$

(vi) $-2/3 + 0$

We know that if any number or fraction is added to zero, the answer will be the same number or fraction.

Hence,

$$= -2/3 + 0$$

$$= -2/3$$

(vii) $-2\frac{1}{3} + 4\frac{3}{5}$

First, we have to convert mixed fractions into improper fractions.

=

$$-2\frac{1}{3} = -7/3$$

=

$$4\frac{3}{5} = 23/5$$

We have, $-7/3 + 23/5$

Take the LCM of the denominators of the given rational numbers.

The LCM of 3 and 5 is 15

Express each of the given rational numbers with the above LCM as the common denominator.

Now,

$$(-7/3) = [(-7 \times 5) / (3 \times 5)] = (-35/15)$$

$$(23/5) = [(23 \times 3) / (5 \times 3)] = (69/15)$$

Then,

$$= (-35/15) + (69/15) \dots [\because \text{denominator is same in both the rational numbers}]$$

$$= (-35 + 69)/15$$

$$= (34/15)$$

Question 2:

Find

$$(i) \frac{7}{24} - \frac{17}{36} \quad (ii) \frac{5}{63} - \left(\frac{-6}{21}\right) \quad (iii) \frac{-6}{13} - \left(\frac{-7}{15}\right)$$

$$(iv) \frac{-3}{8} - \frac{7}{11} \quad (v) -2\frac{1}{9} - 6$$

Answer:

$$(i) \frac{7}{24} - \frac{17}{36}$$

L.C.M of 24 and 36 is 72.

$$\frac{7}{24} - \frac{17}{36} = \frac{7 \times 3}{24 \times 3} - \frac{17 \times 2}{36 \times 2} = \frac{21}{72} - \frac{34}{72} = \frac{21-34}{72} = \frac{-13}{72}$$

$$(ii) \frac{5}{63} - \left(\frac{-6}{21}\right) = \frac{5}{63} + \frac{2}{7}$$

L.C.M of 63 and 7 is 63.

$$\frac{5}{63} + \frac{2}{7} = \frac{5}{63} + \frac{2 \times 9}{7 \times 9} = \frac{5}{63} + \frac{18}{63} = \frac{5+18}{63} = \frac{23}{63}$$

$$(iii) \frac{-6}{13} - \left(\frac{-7}{15}\right) = \frac{-6}{13} + \frac{7}{15}$$

L.C.M of 13 and 15 is 195.

$$\frac{-6}{13} + \frac{7}{15} = \frac{-6 \times 15}{13 \times 15} + \frac{7 \times 13}{15 \times 13} = \frac{-90}{195} + \frac{91}{195} = \frac{-90 + 91}{195} = \frac{1}{195}$$

$$(iv) \quad \frac{-3}{8} - \frac{7}{11}$$

L.C.M of 8 and 11 is 88.

$$\frac{-3}{8} - \frac{7}{11} = \frac{-3 \times 11}{8 \times 11} - \frac{7 \times 8}{11 \times 8} = \frac{-33}{88} - \frac{56}{88} = \frac{-33 - 56}{88} = \frac{-89}{88}$$

$$(v) \quad -2\frac{1}{9} - 6 = -\frac{19}{9} - \frac{6}{1}$$

L.C.M of 9 and 1 is 9.

$$-\frac{19}{9} - \frac{6}{1} = -\frac{19}{9} - \frac{6 \times 9}{1 \times 9} = -\frac{19}{9} - \frac{54}{9} = \frac{-19 - 54}{9} = \frac{-73}{9}$$

Question 3:

Find the product:

$$(i) \quad \frac{9}{2} \times \left(\frac{-7}{4}\right) \quad (ii) \quad \frac{3}{10} \times (-9) \quad (iii) \quad \frac{-6}{5} \times \frac{9}{11}$$

$$(iv) \quad \frac{3}{7} \times \left(\frac{-2}{5}\right) \quad (v) \quad \frac{3}{11} \times \frac{2}{5} \quad (vi) \quad \frac{3}{-5} \times \frac{-5}{3}$$

Answer:

$$(i) \quad \frac{9}{2} \times \left(\frac{-7}{4}\right) = \frac{9 \times (-7)}{2 \times 4} = \frac{-63}{8}$$

$$(ii) \quad \frac{3}{10} \times (-9) = \frac{3}{10} \times \frac{(-9)}{1} = \frac{3 \times (-9)}{10 \times 1} = \frac{-27}{10}$$

$$(iii) \quad \frac{-6}{5} \times \frac{9}{11} = \frac{-6 \times 9}{5 \times 11} = \frac{-54}{55}$$

$$(iv) \frac{3}{7} \times \left(\frac{-2}{5} \right) = \frac{3 \times (-2)}{7 \times 5} = \frac{-6}{35}$$

$$(v) \frac{3}{11} \times \frac{2}{5} = \frac{3 \times 2}{11 \times 5} = \frac{6}{55}$$

$$(vi) \frac{3}{-5} \times \frac{-5}{3} = \frac{3 \times (-5)}{(-5) \times 3} = \frac{-15}{-15} = 1$$

Question 4:

Find the value of:

$$(i) (-4) \div \frac{2}{3} \quad (ii) \frac{-3}{5} \div 2 \quad (iii) \frac{-4}{5} \div (-3)$$

$$(iv) \frac{-1}{8} \div \frac{3}{4} \quad (v) \frac{-2}{13} \div \frac{1}{7} \quad (vi) \frac{-7}{12} \div \left(\frac{-2}{13} \right)$$

$$(vii) \frac{3}{13} \div \left(\frac{-4}{65} \right)$$

Answer:

$$(i) -4 \div \frac{2}{3} = -4 \times \frac{3}{2} = \frac{-12}{2} = -6$$

$$(ii) \frac{-3}{5} \div 2 = \frac{-3}{5} \times \frac{1}{2} = \frac{-3 \times 1}{5 \times 2} = \frac{-3}{10}$$

$$(iii) \frac{-4}{5} \div (-3) = \frac{-4}{5} \times \frac{1}{-3} = \frac{(-4) \times 1}{5 \times (-3)} = \frac{-4}{-15} = \frac{4}{15}$$

$$(iv) \frac{-1}{8} \div \frac{3}{4} = \frac{-1}{8} \times \frac{4}{3} = \frac{-1 \times 4}{8 \times 3} = \frac{-4}{24} = -\frac{1}{6}$$

$$(v) \frac{-2}{13} \div \frac{1}{7} = \frac{-2}{13} \times 7 = \frac{-14}{13}$$

$$(vi) \quad \frac{-7}{12} \div \left(\frac{-2}{13} \right) = \frac{-7}{12} \times \frac{13}{-2} = \frac{(-7) \times 13}{12 \times (-2)} = \frac{-91}{-24} = \frac{91}{24}$$

$$(vii) \quad \frac{3}{13} \div \left(\frac{-4}{65} \right) = \frac{3}{13} \times \frac{65}{-4} = \frac{3 \times 65}{13 \times (-4)} = \frac{195}{-52} = -\frac{15}{4}$$